Mathematics Notes

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1 Sequences

Definition 1.1. A sequence (x_n) in $\mathbb{R}$ converges to $L \in \mathbb{R}$ if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies |x_n - L| < \varepsilon.$$

Theorem 1.2. *Every convergent sequence is bounded.*

Proof. Suppose that (x_n) converges to L. Taking $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies |x_n - L| < 1.$$

Hence, for all $n \geq N$,

$$|x_n| \leq |x_n - L| + |L| < 1 + |L|.$$

The remaining terms $x_1, \ldots, x_{N-1}$ are finitely many, so they are bounded. Therefore, (x_n) is bounded. $\square$

Remark 1.3. The main idea is that convergence controls the tail of the sequence, while the initial finitely many terms are automatically bounded.